Analysis Report for Task 1

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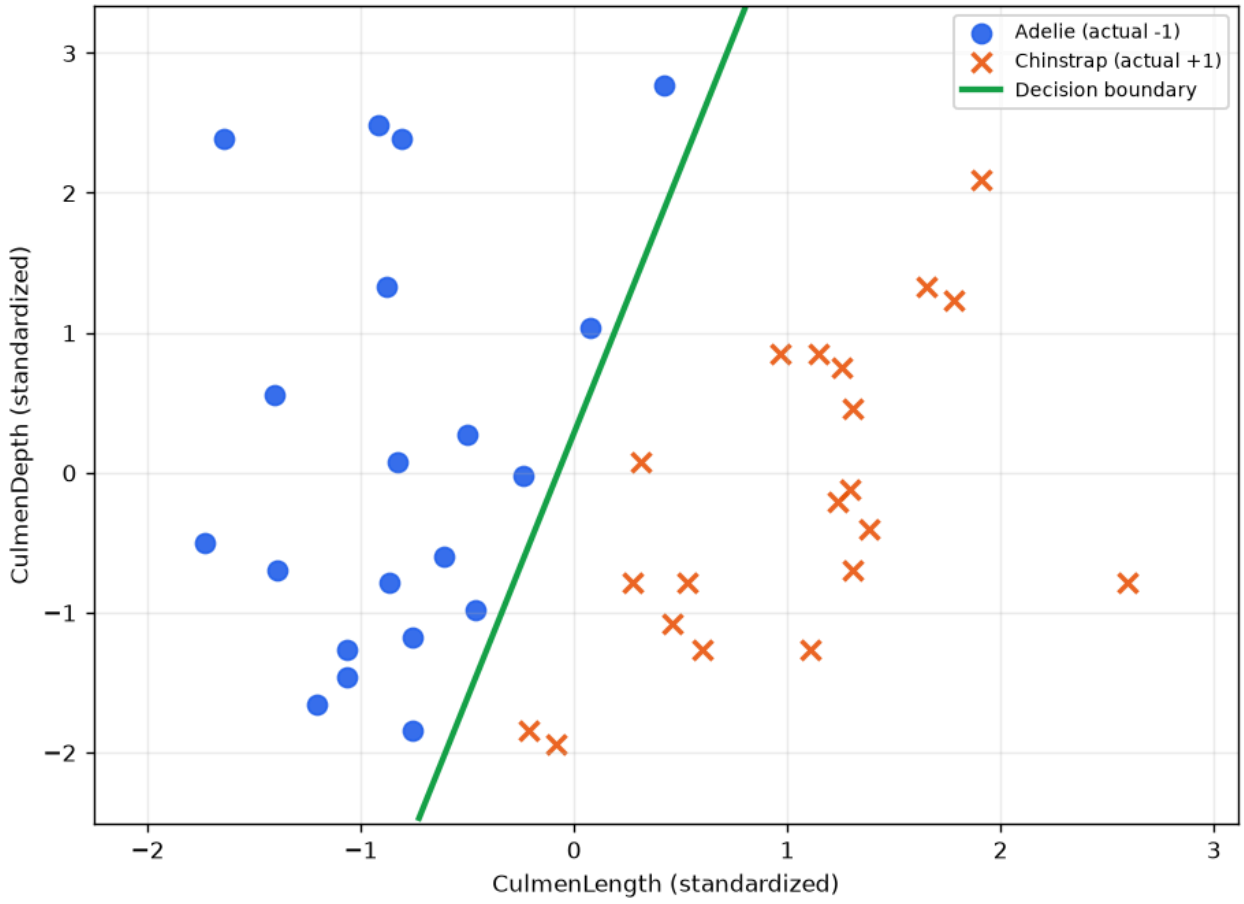
Salma Fathy Saeed

1) Single Layer Perceptron

Experiment 1 : CulmenLength + CulmenDepth
With classes : Adelie vs Chinstrap

Setting	Value
Learning Rate	0.001
Epochs	50
Bias Enabled?	yes
Training Accuracy	100.00%
Testing Accuracy	100.00%

Decision-Boundary Plot



Confusion Matrix

	Predicted: Adelie	Predicted: Chinstrap
Actual: Adelie	**20**	**0**
Actual: Chinstrap	**0**	**20**

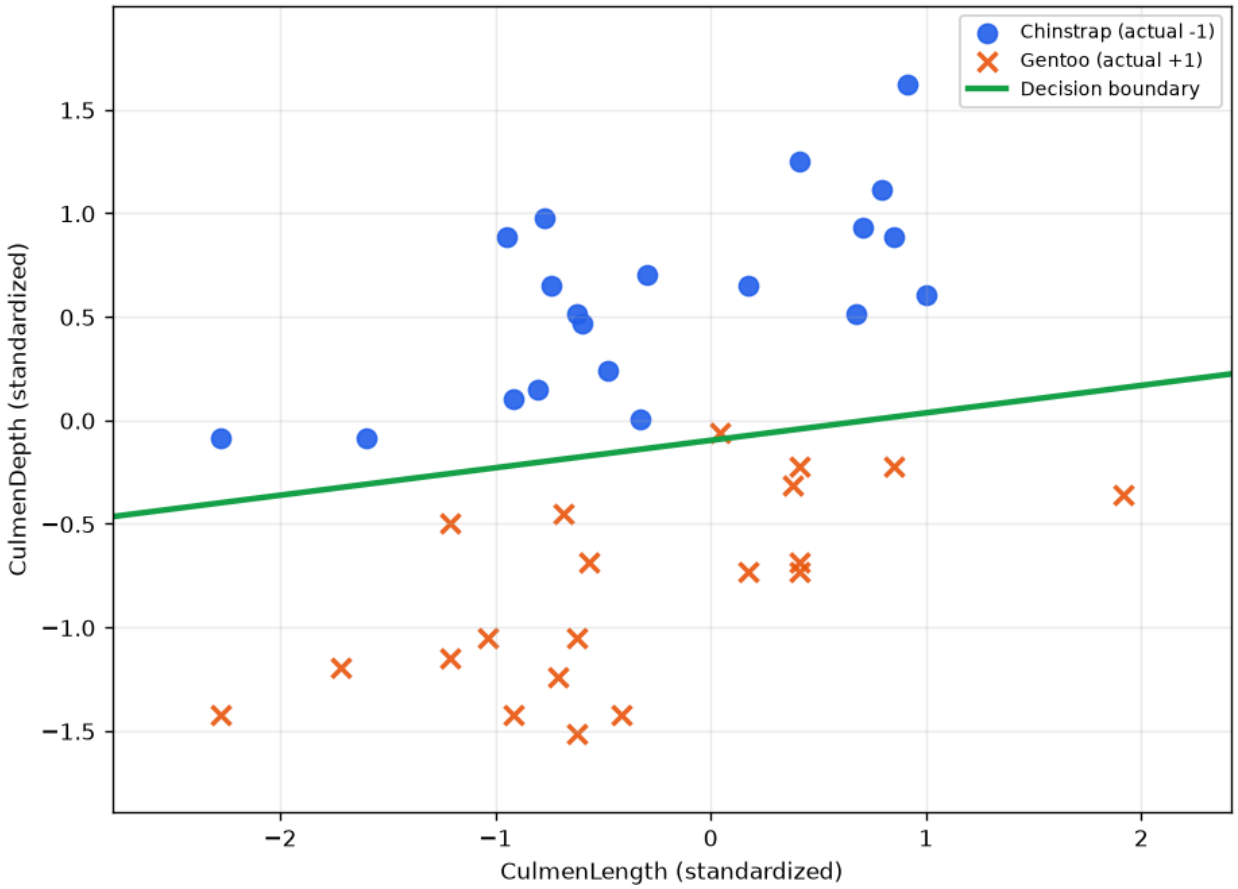
Analysis:

The plot shows that CulmenLength and CulmenDepth separate Adelie and Chinstrap quite well. The decision boundary divides the two classes clearly, with no misclassified points in the test set. Both training and testing accuracy are 100%. The confusion matrix confirms that all 40 test samples were classified correctly.

Experiment 2 : CulmenLength + CulmenDepth
With classes : Gentoo vs Chinstrap

Setting	Value
Learning Rate	0.001
Epochs	50
Bias Enabled?	yes
Training Accuracy	100.00%
Testing Accuracy	97.50%

Decision-Boundary Plot



Confusion Matrix

	Predicted: Chinstrap	Predicted: Gentoo
Actual: Chinstrap	20	0
Actual: Gentoo	1	19

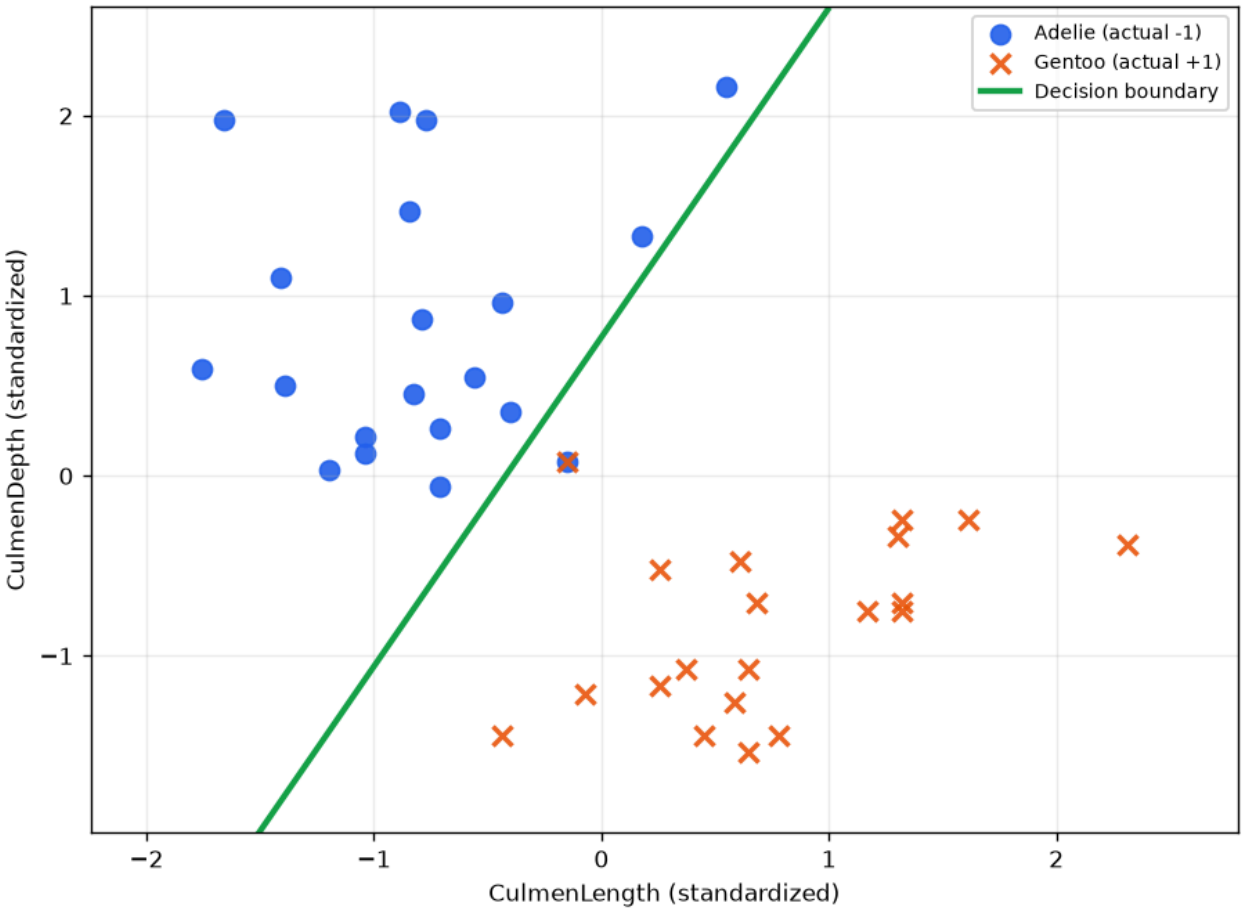
Analysis:

The plot shows that CulmenLength and CulmenDepth separate Chinstrap and Gentoo quite well. The decision boundary classifies most points correctly, with only one Gentoo sample misclassified as Chinstrap. The model achieved 100% training accuracy and 97.5% testing accuracy. The confusion matrix confirms that 39 out of 40 test samples were classified correctly.

Experiment 3 : CulmenLength + CulmenDepth
With classes : Adelie vs Gentoo

Setting	Value
Learning Rate	0.001
Epochs	50
Bias Enabled?	yes
Training Accuracy	100.00%
Testing Accuracy	97.50%

Decision-Boundary Plot



Confusion Matrix

	Predicted: Adelie	Predicted: Gentoo
Actual: Adelie	**19**	**1**
Actual: Gentoo	**0**	**20**

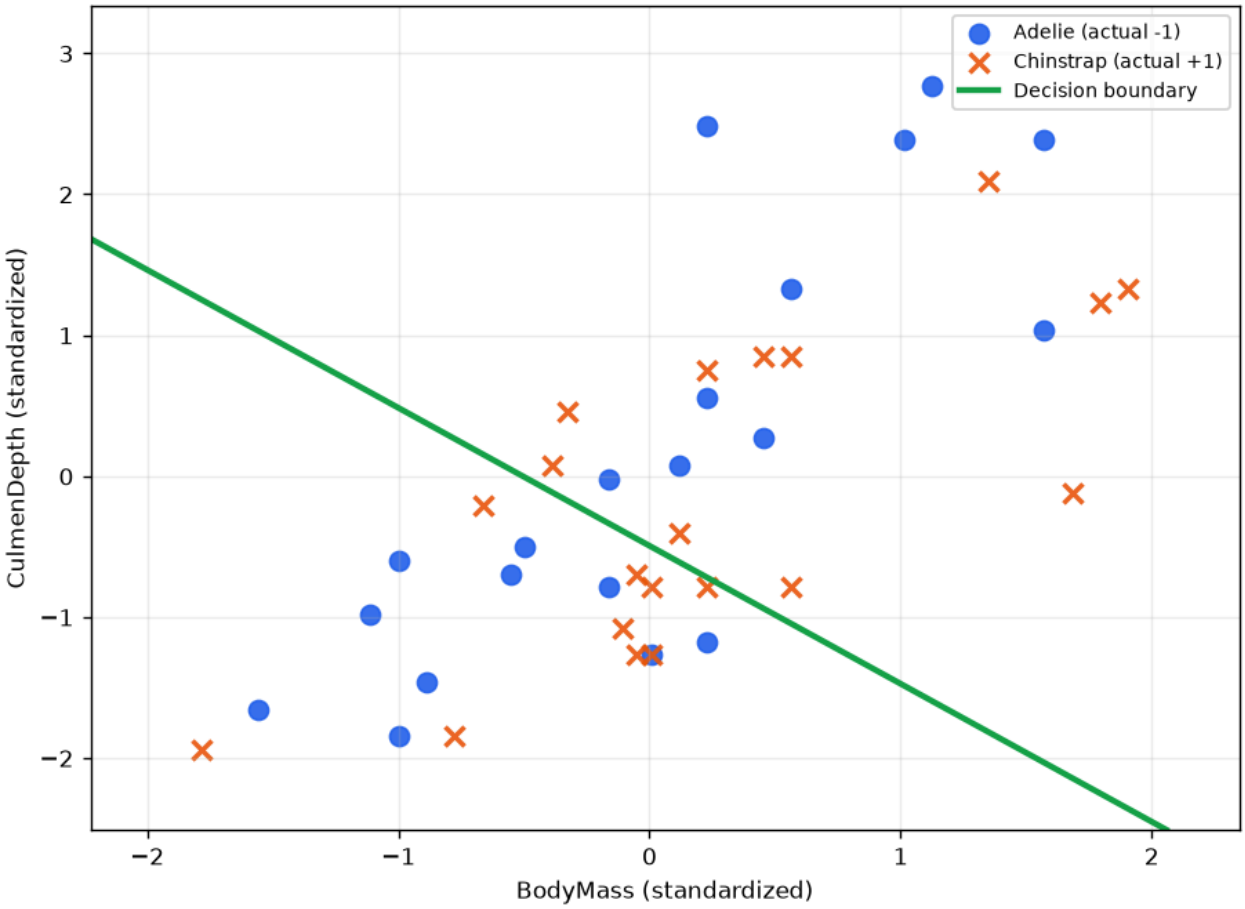
Analysis:

The plot shows that CulmenLength and CulmenDepth separate Adelie and Gentoo quite well. Only one Adelie sample was misclassified as Gentoo, while all Gentoo samples were classified correctly. The model achieved 100% training accuracy and 97.5% testing accuracy. The confusion matrix confirms that 39 out of 40 test samples were classified correctly.

Experiment 4 : BodyMass + CulmenDepth
With classes : Adelie vs ChinStrap

Setting	Value
Learning Rate	0.001
Epochs	50
Bias Enabled?	yes
Training Accuracy	45.00%
Testing Accuracy	52.50%

Decision-Boundary Plot



Confusion Matrix

	Predicted: Adelie	Predicted: Chinstarp
Actual: Adelie	10	10
Actual: Chinstrap	9	11

Analysis:

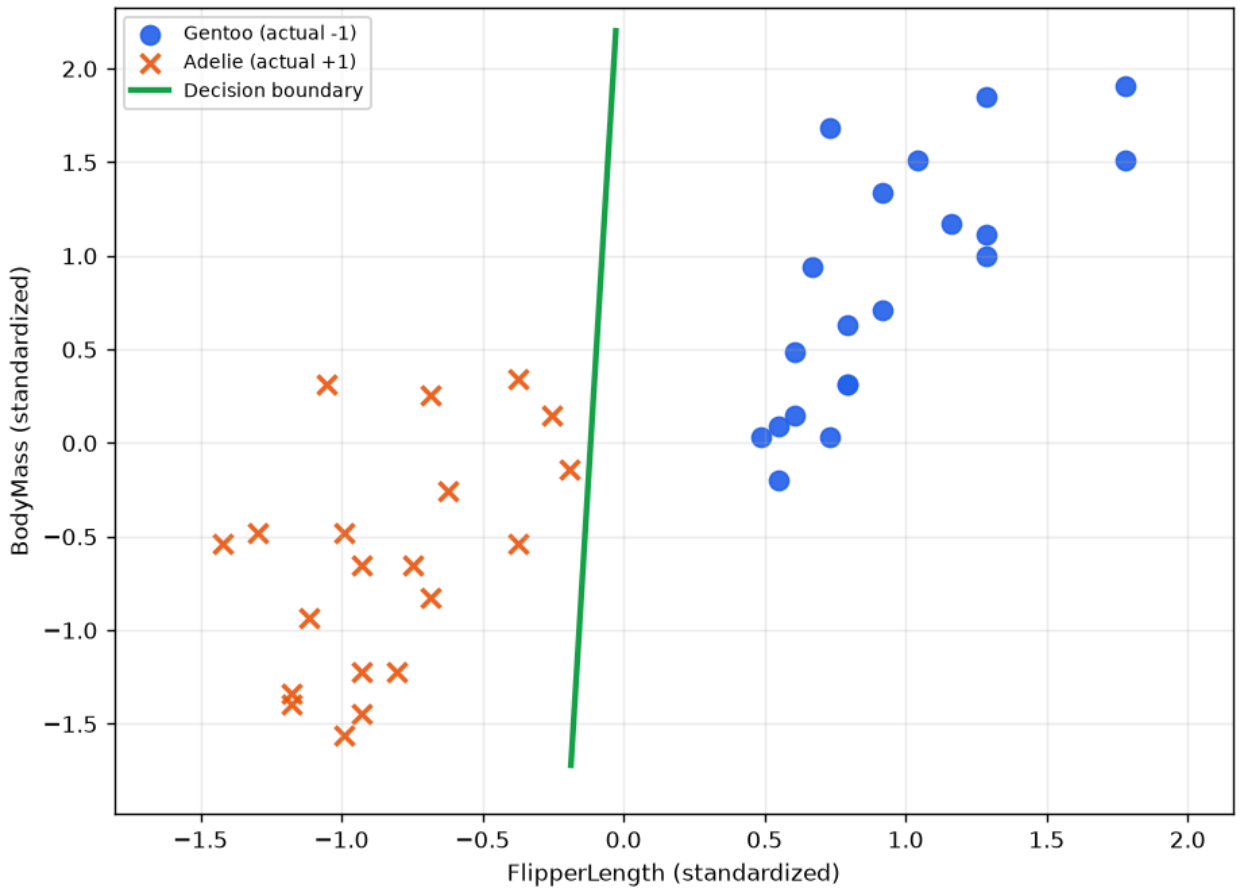
The plot shows that BodyMass and CulmenDepth do not clearly separate Adelie and Chinstrap. Many points overlap around the decision boundary, making classification difficult. The model achieved 45% training accuracy and 52.5% testing accuracy, indicating poor performance with these features. The confusion matrix shows that 10 Adelie and 9 Chinstrap

samples were misclassified. Only 21 out of 40 test samples were classified correctly. This suggests that other feature combinations may be more suitable for distinguishing these two species.

Experiment 5 : FlipperLength + BodyMass
With classes : Gentoo vs Adelie

Setting	Value
Learning Rate	0.01
Epochs	100
Bias Enabled?	yes
Training Accuracy	98.33%
Testing Accuracy	100.00%

Decision-Boundary Plot



Confusion Matrix

	Predicted: Gentoo	Predicted: Adelie
Actual: Gentoo	20	0
Actual: Adelie	0	20

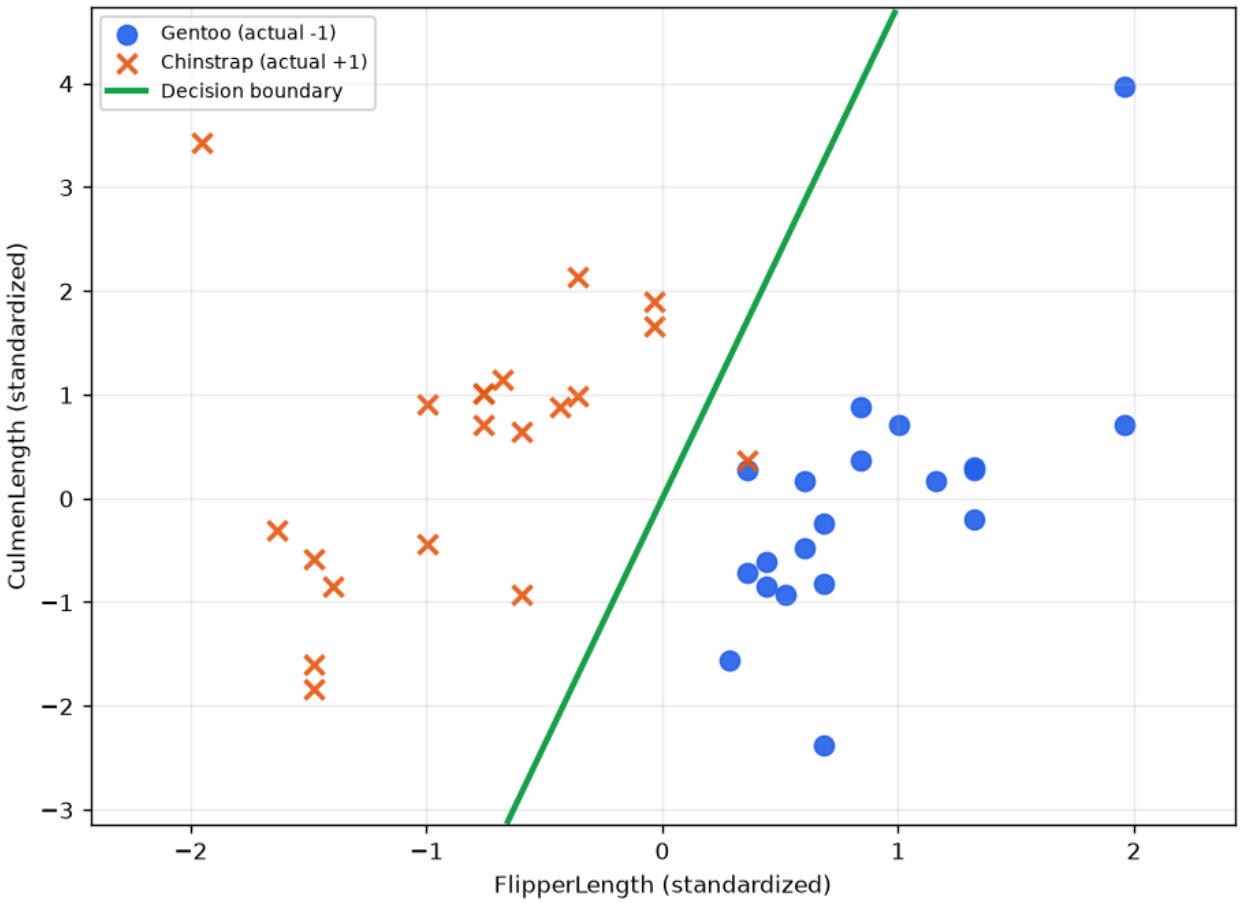
Analysis:

The plot shows that FlipperLength and BodyMass clearly separate Gentoo and Adelie. The decision boundary divides the two classes with no misclassified test samples. The model achieved 98.33% training accuracy and 100% testing accuracy. The confusion matrix confirms that all 40 test samples were classified correctly. These features are highly effective for distinguishing between the two species.

Experiment 6 : FlipperLength + CulmenLength
With classes : Gentoo vs Chinstrap

Setting	Value
Learning Rate	0.001
Epochs	100
Bias Enabled?	no
Training Accuracy	96.67%
Testing Accuracy	97.50%

Decision-Boundary Plot



Confusion Matrix

	Predicted: Gentoo	Predicted: Chinstrap
Actual: Gentoo	20	0
Actual: Chinstrap	1	19

Analysis:

The plot shows that FlipperLength and CulmenLength clearly separate Gentoo and Chinstrap. The decision boundary classifies almost all samples correctly, with only one Chinstrap sample misclassified as Gentoo. The model achieved 96.67% training accuracy and 97.50% testing accuracy. The confusion matrix confirms that 39 out of 40 test samples

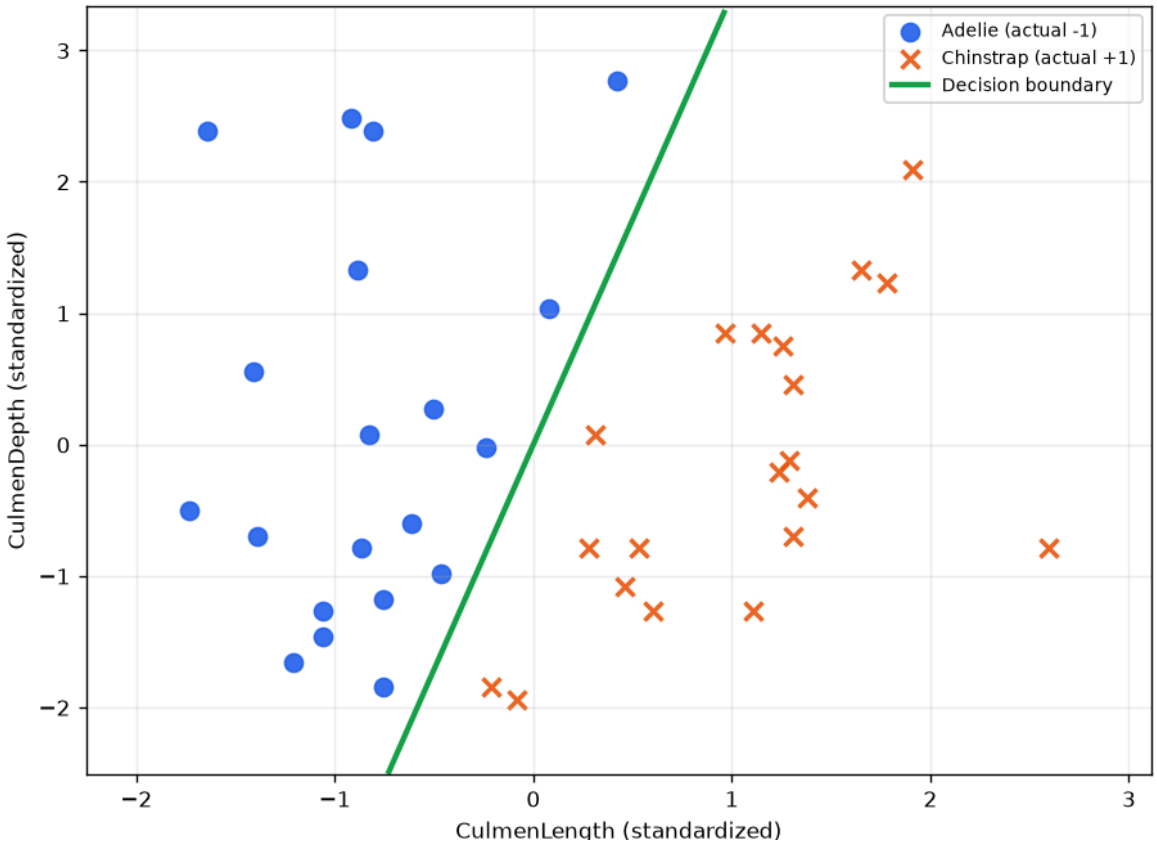
were classified correctly. These features work well for distinguishing between the two species.

2) ADALINE

Experiment 1 : CulmenLength + CulmenDepth
With classes : Adelie vs Chinstrap

Setting	Value
Learning Rate	0.001
Epochs	50
Bias Enabled?	no
MSE threshold	0.01
Training Accuracy	100.00%
Testing Accuracy	100.00%

Decision-Boundary Plot



Confusion Matrix

	Predicted: Adelie	Predicted: Chinstrap
Actual: Adelie	20	0
Actual: Chinstrap	0	20

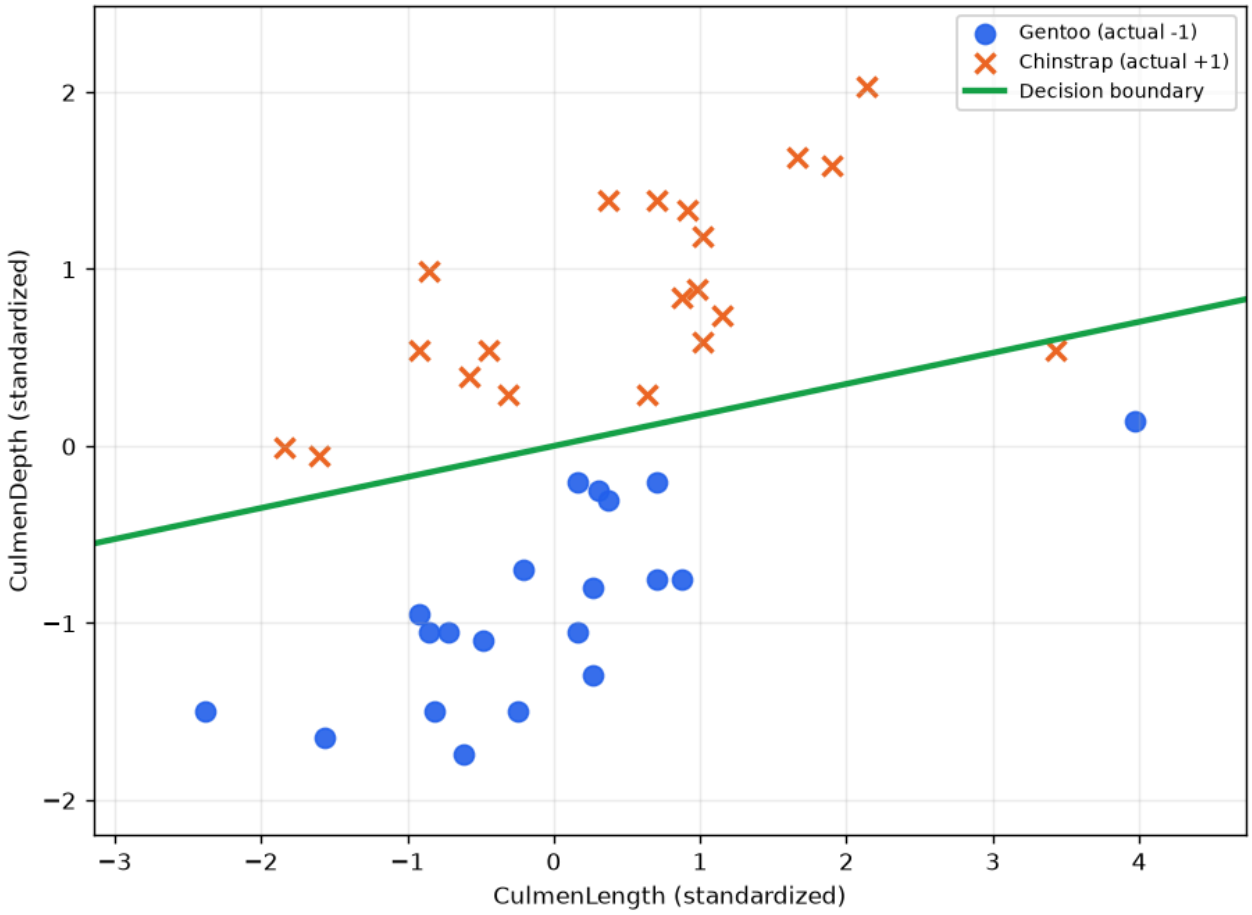
Analysis:

This experiment achieved 100% training and testing accuracy, correctly classifying all 40 test samples.
The decision boundary clearly separates Adelie and Chinstrap, with no misclassified points in the confusion matrix.
This means CulmenLength and CulmenDepth are effective features for distinguishing between these two classes.
The model achieved these results with a learning rate of 0.001 and 50 epochs, without using bias.

Experiment 2 : CulmenLength + CulmenDepth
With classes : Gentoo vs Chinstrap

Setting	Value
Learning Rate	0.001
Epochs	50
Bias Enabled?	no
MSE threshold	0.02
Training Accuracy	100.00%
Testing Accuracy	97.50%

Decision-Boundary Plot



Confusion Matrix

	Predicted: Chinstrap	Predicted: Gentoo
Actual: Chinstrap	20	0
Actual: Gentoo	1	19

Analysis:

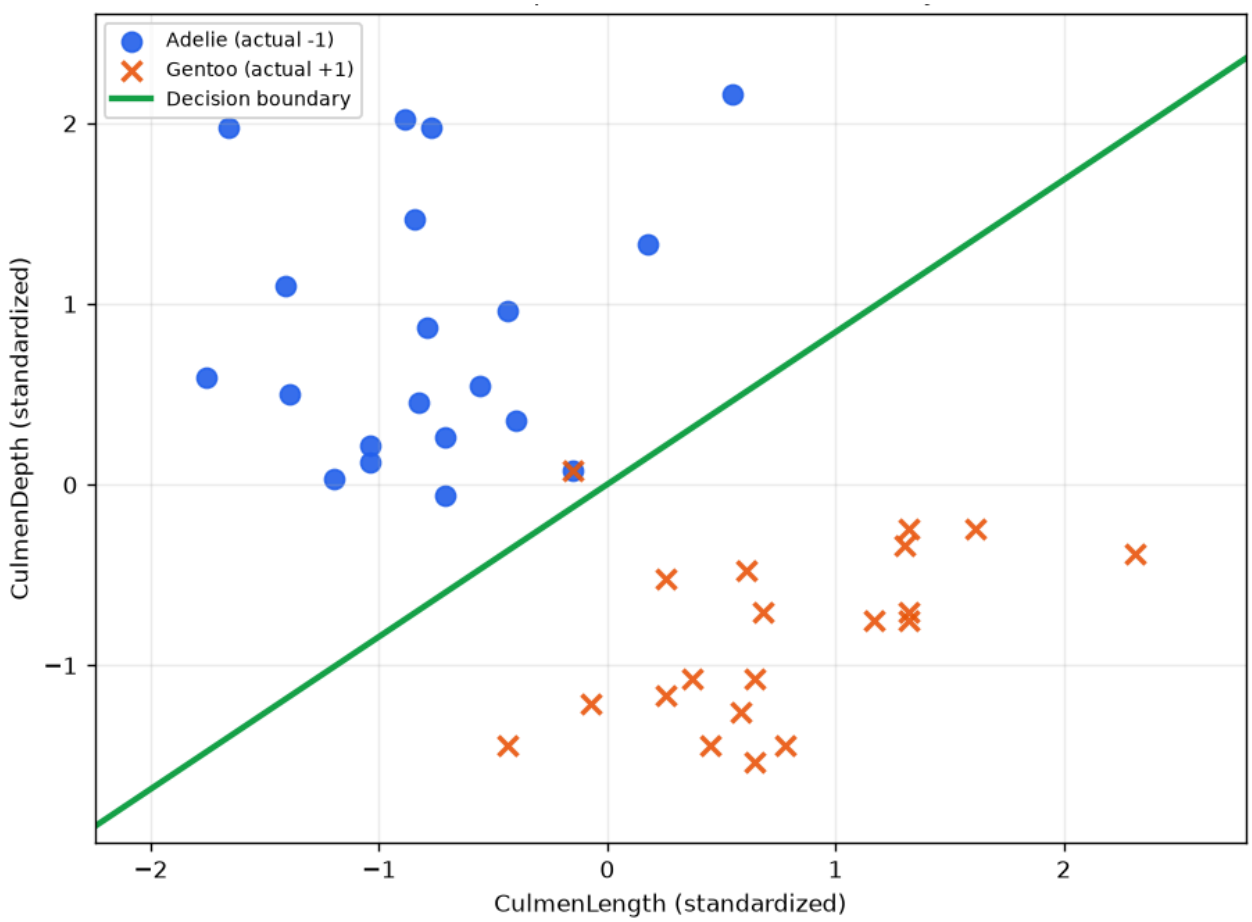
The model achieved 100% training accuracy and 97.50% testing accuracy, meaning it classified almost all unseen samples correctly.

The plot shows that CulmenLength and CulmenDepth separate Gentoo and Chinstrap quite well.
The confusion matrix shows that all 20 Chinstrap samples were classified correctly, while one Gentoo sample was misclassified as Chinstrap.
This means the model made only one mistake out of 40 test samples.
Overall, these features are effective for distinguishing between the two species, even without using bias.

Experiment 3 : CulmenLength + CulmenDepth
With classes : Adelie vs Gentoo

Setting	Value
Learning Rate	0.001
Epochs	50
Bias Enabled?	no
MSE threshold	0.015
Training Accuracy	100.00%
Testing Accuracy	97.50%

Decision-Boundary Plot



Confusion Matrix

	Predicted: Adelie	Predicted: Gentoo
Actual: Adelie	20	0
Actual: Gentoo	1	19

Analysis:

The model achieved 100% training accuracy and 97.50% testing accuracy, showing excellent classification performance.

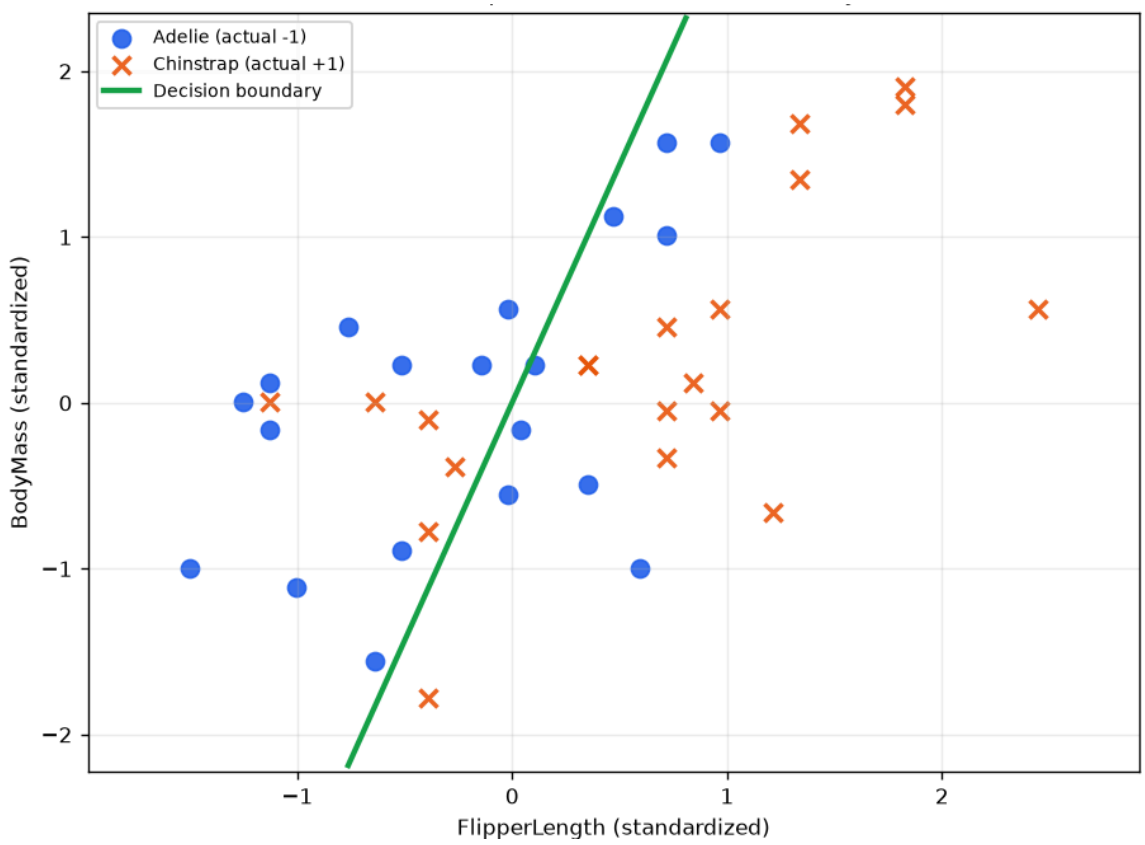
The decision-boundary plot shows that CulmenLength and CulmenDepth clearly separate Adelie and Gentoo.The confusion matrix indicates that all 20 Adelie samples were classified correctly, while one Gentoo sample was misclassified as

Adelie.Therefore, the model correctly classified 39 out of 40 testing samples.These results show that both features are effective for distinguishing the two species, even without bias.

Experiment 4 : FlipperLength + BodyMass
With classes : Adelie vs Chinstrap

Setting	Value
Learning Rate	0.001
Epochs	50
Bias Enabled?	no
MSE threshold	0.015
Training Accuracy	81.67%
Testing Accuracy	65.00%

Decision-Boundary Plot



Confusion Matrix

	Predicted: Adelie	Predicted: Chinstrap
Actual: Adelie	**11**	**9**
Actual: Chinstrap	**5**	**15**

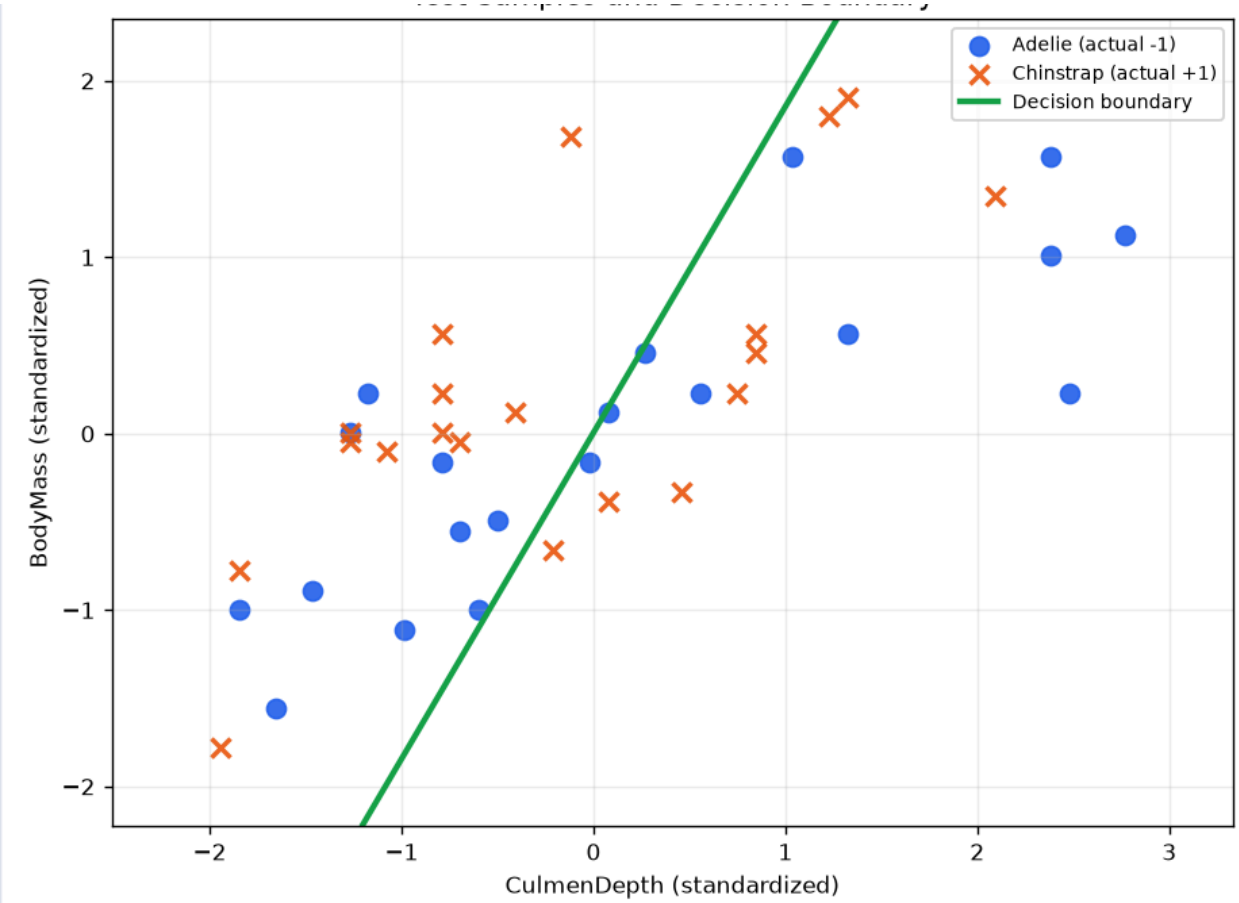
Analysis:

The model achieved 81.67% training accuracy and 65% testing accuracy, showing that its performance is weaker than in previous experiments.
The decision boundary does not clearly separate Adelie and Chinstrap because their feature values overlap.
The confusion matrix shows that 11 Adelie and 15 Chinstrap samples were classified correctly, while 14 samples were misclassified.
This means that FlipperLength and BodyMass are not sufficient for reliable classification in this experiment.
The lower testing accuracy indicates that these features are less effective at distinguishing the two species.

Experiment 5 : CulmenDepth + BodyMass
With classes : Adelie vs Chinstrap

Setting	Value
Learning Rate	0.001
Epochs	200
Bias Enabled?	yes
MSE threshold	0.015
Training Accuracy	58.33%
Testing Accuracy	52.50%

Decision-Boundary Plot



Confusion Matrix

	Predicted: Adelie	Predicted: Chinstrap
Actual: Adelie	10	10
Actual: Chinstrap	9	11

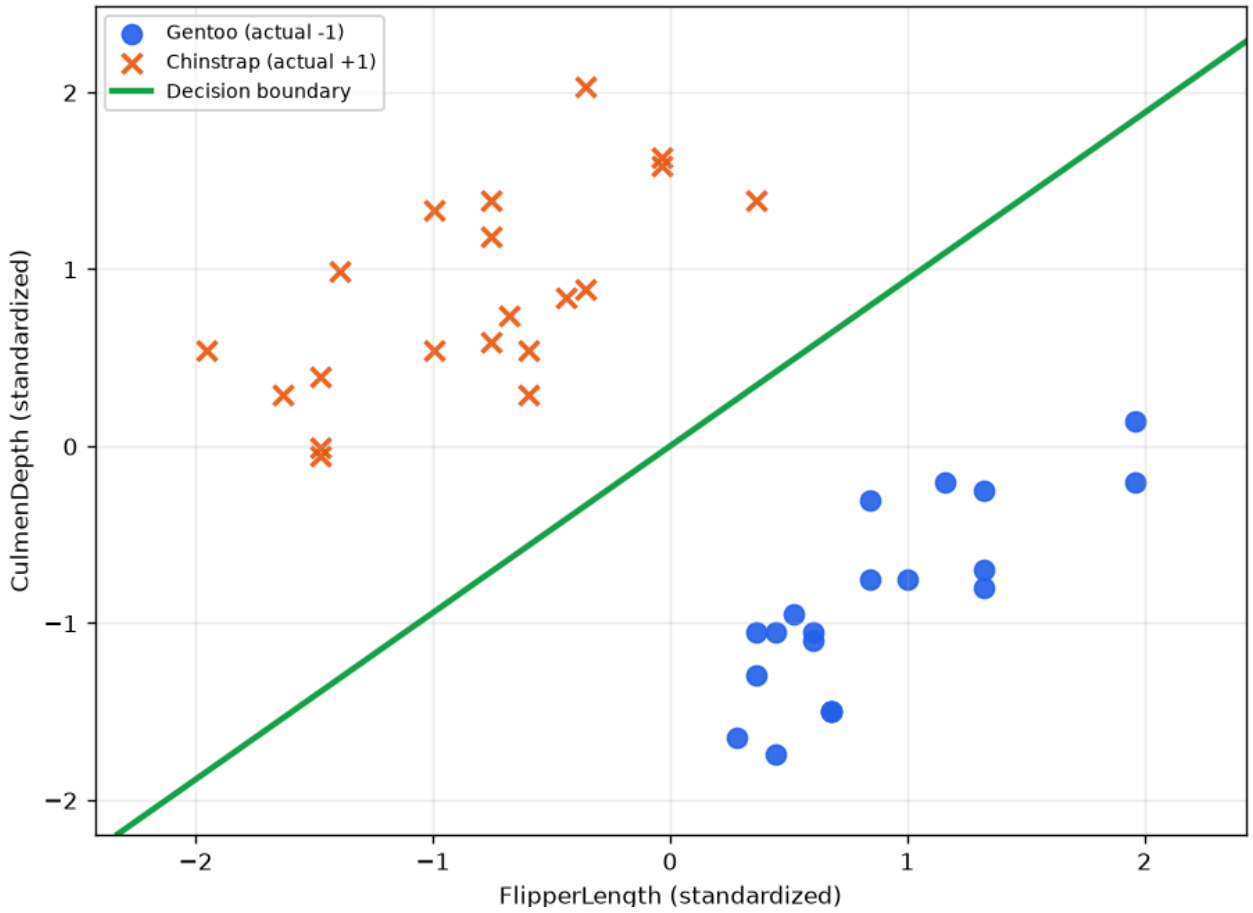
Analysis:

The model achieved 52.50% testing accuracy, showing weak classification performance. The features overlap significantly, causing 19 out of 40 samples to be misclassified. This suggests that CulmenDepth and BodyMass are not effective enough to distinguish Adelie and Chinstrap accurately.

Experiment 6 : FlipperLength + CulmenDepth
With classes : Gentoo vs Chinstrap

Setting	Value
Learning Rate	0.001
Epochs	100
Bias Enabled?	yes
MSE threshold	0.015
Training Accuracy	100.00%
Testing Accuracy	100.00%

Decision-Boundary Plot



Confusion Matrix

	Predicted: Chinstrap	Predicted: Gentoo
Actual: Chinstrap	20	0
Actual: Gentoo	0	20

Analysis:

The model achieved 100% training and testing accuracy, correctly classifying all samples. The decision boundary clearly separates Gentoo and Chinstrap penguins, indicating that FlipperLength and CulmenDepth are highly effective features for this classification task.

3) Overall Findings

The experiments show that classification performance depends strongly on the selected feature pair and the two species being compared. A decision boundary works well when the classes form clearly separated groups; performance drops when their points overlap substantially

Highest reported testing accuracy :

Algorithm	Experiment	Features	Training	Testing
Perceptron	Experiment 1:Adelie vs Chinstrap	CulmenLength + CulmenDepth	**100.00%**	**100.00%**
Perceptron	Experiment 5: Adelie vs Gentoo	FlipperLength + BodyMass	**98.33%**	**100.00%**
ADALINE	Experiment 1:Adelie vs Chinstrap	CulmenLength + CulmenDepth	**100.00%**	**100.00%**
ADALINE	Experiment 6: Gentoo vs Chinstrap	FlipperLength + CulmenDepth	**100.00%**	**100.00%**

The highest testing accuracy is 100%, achieved in the experiments listed above. These results indicate that
CulmenLength + CulmenDepth is especially effective for separating Adelie and Chinstrap in the reported runs.
FlipperLength + CulmenDepth also separates Gentoo and Chinstrap very effectively in the ADALINE experiment.
FlipperLength + BodyMass gives excellent separation for Gentoo and Adelie in the Perceptron experiment.